Advanced Trigonometry

ANSWER KEY

Compound angle formulas

1 Expand $\sin(x+30^\circ)$ in terms of $\sin x$ and $\cos x$, using exact values for $\sin 30^\circ$ and $\cos 30^\circ$.
$\sin(x+30^\circ) = \sin x\cos 30^\circ + \cos x\sin 30^\circ = \dfrac{\sqrt{3}}{2}\sin x + \dfrac{1}{2}\cos x$.

2 Given $\sin A = \dfrac{3}{5}$ and $\cos B = \dfrac{12}{13}$, with A and B both acute, find $\cos(A+B)$.
Since A, B are acute, $\cos A = \dfrac{4}{5}$ and $\sin B = \dfrac{5}{13}$.
$\cos(A+B) = \cos A\cos B - \sin A\sin B = \dfrac{4}{5}\cdot\dfrac{12}{13} - \dfrac{3}{5}\cdot\dfrac{5}{13} = \dfrac{48}{65} - \dfrac{15}{65} = \dfrac{33}{65}$.

3 Use the compound-angle formula for tangent to find the exact value of $\tan 75^\circ$, using $75^\circ = 45^\circ + 30^\circ$
$\tan 75^\circ = \dfrac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ} = \dfrac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}} = \dfrac{\sqrt{3}+1}{\sqrt{3}-1}$. Rationalizing:
$\dfrac{(\sqrt{3}+1)^2}{(\sqrt{3}-1)(\sqrt{3}+1)} = \dfrac{4+2\sqrt{3}}{2} = 2+\sqrt{3}$.

4 Use the compound-angle formula for cosine to find the exact value of $\cos 15^\circ$, using $15^\circ = 45^\circ - 30^\circ$.
$\cos 15^\circ = \cos 45^\circ\cos 30^\circ + \sin 45^\circ\sin 30^\circ = \dfrac{\sqrt{2}}{2}\cdot\dfrac{\sqrt{3}}{2} + \dfrac{\sqrt{2}}{2}\cdot\dfrac{1}{2} = \dfrac{\sqrt{6}+\sqrt{2}}{4}$.

Double angle formulas

5 Given $\sin\theta = \dfrac{4}{5}$ with θ acute, find $\sin 2\theta$ and $\cos 2\theta$.
Since θ is acute, $\cos\theta = \dfrac{3}{5}$. $\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\dfrac{4}{5}\right)\left(\dfrac{3}{5}\right) = \dfrac{24}{25}$.
$\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2\left(\dfrac{16}{25}\right) = -\dfrac{7}{25}$.

6 Given $\cos\theta = -\dfrac{1}{3}$ with $90^\circ < \theta < 180^\circ$, find $\cos 2\theta$ and state which quadrant 2θ lies in.
$\cos 2\theta = 2\cos^2\theta - 1 = 2\left(\dfrac{1}{9}\right) - 1 = -\dfrac{7}{9}$. Since $90^\circ < \theta < 180^\circ$, 2θ lies between 180° and 360°; combined with $\cos 2\theta < 0$ (which requires an angle between 90° and 270°), 2θ must lie in the third quadrant, i.e. $180^\circ < 2\theta < 270^\circ$.

7 Given $\tan\theta = \dfrac{1}{2}$ with θ acute, find $\tan 2\theta$, using $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$.
$\tan 2\theta = \dfrac{2\left(\frac{1}{2}\right)}{1-\left(\frac{1}{2}\right)^2} = \dfrac{1}{1-\frac{1}{4}} = \dfrac{1}{3/4} = \dfrac{4}{3}$.

Solving equations using double angle formulas

8 Solve $\cos 2\theta = \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.
$1 - 2\sin^2\theta = \sin\theta \Rightarrow 2\sin^2\theta + \sin\theta - 1 = 0 \Rightarrow (2\sin\theta - 1)(\sin\theta + 1) = 0$. So $\sin\theta = \dfrac{1}{2}$, giving $\theta = 30^\circ$ or 150°; or $\sin\theta = -1$, giving $\theta = 270^\circ$.

9 Solve $\sin 2\theta = \cos\theta$ for $0^\circ \le \theta < 360^\circ$, using $\sin 2\theta = 2\sin\theta\cos\theta$.
$2\sin\theta\cos\theta - \cos\theta = 0 \Rightarrow \cos\theta(2\sin\theta - 1) = 0$. So $\cos\theta = 0$, giving $\theta = 90^\circ$ or 270°; or $\sin\theta = \dfrac{1}{2}$, giving $\theta = 30^\circ$ or 150°.

The R sin(θ+α) form

10 Express $3\sin\theta + 4\cos\theta$ in the form $R\sin(\theta+\alpha)$, with $R>0$ and $0^\circ < \alpha < 90^\circ$, giving α to the nearest degree.

$R = \sqrt{3^2+4^2} = 5$. $\tan\alpha = \frac{4}{3} \Rightarrow \alpha \approx 53^\circ$. So $3\sin\theta + 4\cos\theta = 5\sin(\theta+53^\circ)$.

11 Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta-\alpha)$, giving exact values of R and α.

$R = \sqrt{(\sqrt{3})^2+1^2} = \sqrt{4} = 2$. $\tan\alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ$. So $\sqrt{3}\sin\theta - \cos\theta = 2\sin(\theta-30^\circ)$.

Using the R-form to find extrema and solve equations

12 The graph shows $y = 5\sin(x+53^\circ)$, the R-form of $3\sin x + 4\cos x$ found earlier. State the maximum and minimum values of $3\sin\theta + 4\cos\theta$, and the value of θ (to the nearest degree, $0^\circ \le \theta < 360^\circ$) at which the maximum occurs.

Since $-1 \le \sin(\theta+53^\circ) \le 1$, the maximum value is $R=5$ and the minimum is -5. The maximum occurs when $\theta + 53^\circ = 90^\circ$, i.e. $\theta = 37^\circ$.

13 Solve $\sqrt{3}\sin\theta - \cos\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using the R-form $2\sin(\theta-30^\circ)$ found earlier.

$2\sin(\theta-30^\circ) = 1 \Rightarrow \sin(\theta-30^\circ) = \frac{1}{2}$. For $\theta - 30^\circ$ in the range $-30^\circ \le \theta - 30^\circ < 330^\circ$:
$\theta - 30^\circ = 30^\circ$ or 150°, giving $\theta = 60^\circ$ or $\theta = 180^\circ$.

Proving trigonometric identities

14 Prove that $\frac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$.

Using $\sin 2\theta = 2\sin\theta\cos\theta$ and $\cos 2\theta = 2\cos^2\theta - 1$:

$$\frac{2\sin\theta\cos\theta}{1+(2\cos^2\theta-1)} = \frac{2\sin\theta\cos\theta}{2\cos^2\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta.$$

15 Prove that $\cos^4\theta - \sin^4\theta = \cos 2\theta$.

Factor as a difference of squares: $\cos^4\theta - \sin^4\theta = (\cos^2\theta - \sin^2\theta)(\cos^2\theta + \sin^2\theta)$. Since $\cos^2\theta + \sin^2\theta = 1$, this simplifies to $\cos^2\theta - \sin^2\theta$, which is exactly $\cos 2\theta$.

16 Prove that $(\sin\theta + \cos\theta)^2 = 1 + \sin 2\theta$.

Expand the left side:
$\sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta = (\sin^2\theta + \cos^2\theta) + 2\sin\theta\cos\theta = 1 + 2\sin\theta\cos\theta = 1 + \sin 2\theta$.

Solving equations that combine several identities

17 Solve $2\cos 2\theta + 3\sin\theta = 1$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$, giving answers to one decimal place where needed.

$2(1-2\sin^2\theta) + 3\sin\theta = 1 \Rightarrow 2 - 4\sin^2\theta + 3\sin\theta - 1 = 0 \Rightarrow 4\sin^2\theta - 3\sin\theta - 1 = 0 \Rightarrow (4\sin\theta + 1)(\sin\theta - 1) = 0$
So $\sin\theta = 1$, giving $\theta = 90^\circ$; or $\sin\theta = -\frac{1}{4}$, giving $\theta \approx 194.5^\circ$ or $\theta \approx 345.5^\circ$.

18 Solve $\cos 2\theta = 1 - \sin\theta$ for $0^\circ \le \theta < 360^\circ$, using $\cos 2\theta = 1 - 2\sin^2\theta$.

$1 - 2\sin^2\theta = 1 - \sin\theta \Rightarrow -2\sin^2\theta + \sin\theta = 0 \Rightarrow \sin\theta(1 - 2\sin\theta) = 0$. So $\sin\theta = 0$, giving $\theta = 0^\circ$ or 180°; or $\sin\theta = \frac{1}{2}$, giving $\theta = 30^\circ$ or 150°.

A well-designed word problem: alternating current

19 This question has three parts. The voltage in an AC circuit is modeled by $V(t) = 6\sin(100\pi t) + 8\cos(100\pi t)$ volts, where t is time in seconds. (a) Express $V(t)$ in the form $R\sin(100\pi t + \alpha)$, giving R exactly and α in radians to four decimal places. (b) State the maximum voltage reached, and the smallest positive value of t (in seconds, to four significant figures) at which it occurs. (c) Find the two values of t in $0 \le t < 0.02$ at which $V(t) = 5$, to four significant figures.

(a) $R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. $\alpha = \arctan\left(\frac{8}{6}\right) \approx 0.9273$ rad. So $V(t) = 10\sin(100\pi t + 0.9273)$. (b) The maximum voltage is $R = 10$ volts, occurring when $100\pi t + 0.9273 = \frac{\pi}{2}$, i.e. $t = \frac{1.5708 - 0.9273}{100\pi} \approx 0.002049$ s. (c) Solve $10\sin(100\pi t + 0.9273) = 5 \Rightarrow \sin(100\pi t + 0.9273) = 0.5$. Using $100\pi t + 0.9273 = \frac{5\pi}{6} \approx 2.6180$: $t = \frac{2.6180 - 0.9273}{100\pi} \approx 0.005382$ s. Using $100\pi t + 0.9273 = \frac{\pi}{6} + 2\pi \approx 6.8068$: $t = \frac{6.8068 - 0.9273}{100\pi} \approx 0.01872$ s.